Vehicle Parking Management System

Code Coverage Report

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Executed 8 November 2026 on commit 39e0e56. 748 of 748 Jest tests passed in 39 suites.

1. Scope and method

Coverage was measured with Jest's built-in Istanbul instrumentation while the full unit and integration suite ran (npm run test:coverage). It covers all server code in server/src except the start-up file server/src/index.js, which only opens the network port. The React web client is not instrumented; it is exercised by the browser tests reported in the System Validation Report.

- Statement coverage: share of executable statements run at least once.
- Line coverage: share of source lines with code that ran.
- Function coverage: share of functions called at least once.
- Branch coverage is reported in detail in the Branch Coverage Report.

2. Summary

Metric	Covered	Total	Coverage
Statements	1329	1343	98.95%
Lines	1216	1225	99.26%
Functions	219	221	99.09%
Branches	626	675	92.74%

71 files measured; 62 have 100% statement coverage.

3. Coverage by file

File (server/src/)	Statements	Lines	Functions	Uncovered lines
app.js	98.14%	100%	75.00%	71
config.js	90.90%	100%	100%	11
db.js	100%	100%	100%	-
errors.js	100%	100%	100%	-
middleware/authenticate.js	100%	100%	100%	-
middleware/errorHandler.js	93.75%	100%	100%	6
middleware/requireRole.js	100%	100%	100%	-
modules/admin/admin.controller.js	100%	100%	100%	-
modules/admin/admin.repository.js	100%	100%	100%	-
modules/admin/admin.routes.js	100%	100%	100%	-
modules/admin/admin.service.js	93.87%	93.33%	100%	23, 45, 73
modules/admin/admin.validation.js	97.05%	100%	100%	32
modules/audit/audit.controller.js	100%	100%	100%	-
modules/audit/audit.repository.js	100%	100%	100%	-
modules/audit/audit.routes.js	100%	100%	100%	-
modules/audit/audit.service.js	100%	100%	100%	-
modules/audit/audit.validation.js	100%	100%	100%	-
modules/audit/auditActions.js	100%	100%	100%	-
modules/history/history.controller.js	100%	100%	100%	-
modules/history/history.repository.js	100%	100%	100%	-
modules/history/history.routes.js	100%	100%	100%	-

File (server/src/)	Statements	Lines	Functions	Uncovered lines
modules/history/history.service.js	100%	100%	100%	-
modules/history/history.validation.js	100%	100%	100%	-
modules/parking/parking.controller.js	100%	100%	100%	-
modules/parking/parking.repository.js	100%	100%	100%	-
modules/parking/parking.routes.js	100%	100%	100%	-
modules/parking/parking.service.js	100%	100%	100%	-
modules/parking/parking.validation.js	100%	100%	100%	-
modules/parking/sessionStatuses.js	100%	100%	100%	-
modules/payments/payments.controller.js	100%	100%	100%	-
modules/payments/payments.repository.js	100%	100%	100%	-
modules/payments/payments.routes.js	100%	100%	100%	-
modules/payments/payments.service.js	100%	100%	100%	-
modules/payments/payments.validation.js	100%	100%	100%	-
modules/rates/fees.js	100%	100%	100%	-
modules/rates/rates.controller.js	100%	100%	100%	-
modules/rates/rates.repository.js	100%	100%	100%	-
modules/rates/rates.routes.js	100%	100%	100%	-
modules/rates/rates.service.js	100%	100%	100%	-
modules/rates/rates.validation.js	100%	100%	100%	-
modules/reports/reports.controller.js	100%	100%	100%	-
modules/reports/reports.repository.js	100%	100%	100%	-
modules/reports/reports.routes.js	100%	100%	100%	-
modules/reports/reports.service.js	100%	100%	100%	-
modules/reports/reports.validation.js	100%	100%	100%	-
modules/reservations/reservations.controller.js	100%	100%	100%	-
modules/reservations/reservations.repository.js	100%	100%	100%	-
modules/reservations/reservations.routes.js	100%	100%	100%	-
modules/reservations/reservations.service.js	95.65%	95.65%	100%	45
modules/reservations/reservations.validation.js	100%	100%	100%	-
modules/slots/slotStatuses.js	100%	100%	100%	-
modules/slots/slotTypes.js	100%	100%	100%	-
modules/slots/slots.controller.js	100%	100%	100%	-
modules/slots/slots.repository.js	100%	100%	100%	-
modules/slots/slots.routes.js	100%	100%	100%	-
modules/slots/slots.service.js	92.00%	90.47%	83.33%	6, 31
modules/slots/slots.validation.js	100%	100%	100%	-
modules/users/password.js	100%	100%	100%	-
modules/users/roles.js	100%	100%	100%	-
modules/users/sessions.repository.js	100%	100%	100%	-
modules/users/users.controller.js	100%	100%	100%	-
modules/users/users.repository.js	100%	100%	100%	-
modules/users/users.routes.js	100%	100%	100%	-
modules/users/users.service.js	98.00%	100%	100%	86
modules/users/users.validation.js	100%	100%	100%	-
modules/vehicles/vehicles.controller.js	100%	100%	100%	-
modules/vehicles/vehicles.repository.js	100%	100%	100%	-
modules/vehicles/vehicles.routes.js	100%	100%	100%	-
modules/vehicles/vehicles.service.js	90.90%	90.62%	100%	13, 53, 66
modules/vehicles/vehicles.validation.js	100%	100%	100%	-
sessionCookie.js	100%	100%	100%	-
Total	**98.95%**	**99.26%**	**99.09%**	

4. Code not executed

The few statements not executed are the same code paths explained branch by branch in the Branch Coverage Report: re-throwing an unexpected database error, environment defaults in config.js, serving the built client, and guards for states the database constraints rule out. The two functions never called are not used by any route:

- app.js: (anonymous_3) (line 71)
- modules/slots/slots.service.js: list (line 5)

5. How to reproduce

npm run test:coverage then open coverage/index.html for the line-by-line view. The HTML report from this run is included as Coverage_HTML/index.html.